

Associations Between Sociodemographic Variables, Social Determinants of Health, and Diabetes: Findings From a Congregational Health Needs Assessment

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Abstract

Purpose: To examine associations between sociodemographic variables, social determinants of health (SDOHs) and diabetes using health needs assessment data.

Design: Cross-sectional study.

Setting: Faith-based communities in the Mid-South U.S.

Sample: Of the 378 churches, 92 participated in the study (24% response rate); $N = 828$ church leaders and members completed the survey.

Measure: The Mid-South Congregational Health Survey assessed perceived health-related needs of congregations and the communities they serve.

Analysis: Generalized linear mixed modeling examined the associations between sociodemographic variables (age, sex, race/ethnicity, educational level), SDOHs (affordable healthcare, healthy food, employment), and diabetes.

Results: Individuals with less education had lower odds of reporting all SDOHs as health needs compared to individuals with more education ($OR_{range} = .59-.63$). Men had lower odds of reporting diabetes as a health need or concern compared to women ($OR = .70$; 95% $CI = .50, .97$). African Americans had greater odds of reporting diabetes as a health need compared to individuals in the 'Other' race/ethnicity category ($OR = 3.91$; 95% $CI = 2.20, 6.94$). Individuals who reported affordable healthcare ($OR = 2.54$; 95% $CI = 1.73, 3.72$), healthy food ($OR = 2.24$; 95% $CI = 1.55, 3.24$), and employment ($OR = 3.33$; 95% $CI = 2.29, 4.84$) as health needs had greater odds of reporting diabetes as a health need compared to those who did not report these SDOHs as needs.

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Conclusions: Future studies should evaluate strategies to merge healthcare and faith-based organizations' efforts to address SDOHs impacting diabetes.

Keywords

faith-based organization, community-based participatory research, health needs assessment, health priorities, chronic disease

Purpose

In the United States (U.S.), diabetes (type 1, type 2, and gestational) affects over 37 million people and disproportionately burdens certain populations.¹ The highest prevalence of diabetes in the U.S. is observed among American Indians/Alaska Natives (15%), followed by Hispanics (13%), African Americans (12%), Asians (9%), and Whites (8%).¹ Men report a diagnosis of diabetes more often than women,¹ and the highest prevalence of diabetes is observed in adults between the ages of 45-64 years old.¹ However, diabetes rates in young adults have continued to increase in recent years.² The prevalence of diabetes also varies geographically. The Mid-South (ie, western Tennessee, northern Mississippi) has been shown to have an increased prevalence of chronic diseases including diabetes compared to the rest of the U.S. population, with African Americans and socioeconomically disadvantaged populations experiencing the highest rates of diabetes in this region.³ While genetics and lifestyle play an important role in the prevalence of diabetes,^{4,5} social and environmental barriers significantly impact diabetes disparities.⁶⁻⁹

Social determinants of health (SDOHs) are defined as the conditions in which we are born, grow, age, and work that shape the conditions of our daily life (eg, healthcare access, healthy food access, employment)⁹ and have been shown to contribute to intergroup differences in health outcomes,¹⁰ including diabetes risk and prevalence.^{6,7} For example, access to affordable healthcare and healthy foods are associated with the risk of developing diabetes as well as diabetes-related complications.^{11,12} Understanding SDOHs that contribute to inequalities is needed to reduce individuals being disproportionately affected by diabetes.

Multisector community partnerships are a longstanding approach to improving health and addressing SDOHs.¹³ These partnerships include healthcare systems, academic institutions, faith-based organizations and other community-based institutions who collaborate to improve the health of their communities.^{7,13} Multisector partnerships have been shown to build community capacity for addressing SDOHs through developing new or enhanced data systems, leveraging health-related resources, and engaging community members around important issues.¹³ Community-based participatory research (CBPR) is an approach that equitably involves partners in designing and implementing efforts to address community-level health concerns.¹⁴ Studies using CBPR have demonstrated improvements in diabetes outcomes (eg, hemoglobin

A1c, blood pressure, cholesterol)¹⁵ while other studies have identified resources, community assets, and build multilevel programs to address SDOHs.^{16,17}

Health needs assessments are also useful in assessing community-level health concerns and designing strategies and policies to address identified needs.¹⁸ These assessments can assist organizations and partnerships in prioritizing resources, especially in areas where resources are limited.¹⁹⁻²¹ CBPR-driven health needs assessments developed and implemented in partnership with faith-based organizations have demonstrated success in identifying issues relevant to church and community members, including diabetes.²²⁻²⁴ Despite previous successes,^{22,23} needs related to SDOHs are less often assessed or identified as top health concerns. Studies linking SDOHs and diabetes using CBPR-driven health needs assessment methods are also limited.

This study used data from a CBPR-driven congregational health needs assessment implemented in the greater Memphis, TN area to examine congregation members' perceptions of needs related to diabetes and prominent SDOHs in their community (ie, affordable healthcare, healthy food, employment).²⁴⁻²⁷ The first aim examined odds of sociodemographic categories reporting diabetes and SDOHs as health needs in their community. Sociodemographic variables selected were based on those noted to indicate diabetes inequalities in other research (ie, age, sex, race and ethnicity, educational level).^{1,2} The second aim examined the odds of selecting diabetes as a health need while also selecting community SDOHs as health needs. Examining these relationships has the potential to inform the development of diabetes programs for congregations as well as strategies congregations and communities can use to address diabetes inequalities.

Methods

Design

The study used a cross-sectional design. Data were collected between February 2019 and February 2020 by two faith-based networks (ie, Methodist Le Bonheur's Congregational Health Network, Church Health's Congregational Health Promoters Program) serving over 400 predominantly African American congregations in the Mid-South (ie, western Tennessee, northern Mississippi), an area experiencing high poverty rates²⁸ as well as inequalities in diabetes prevalence.²⁹ The study design and implementation have been described

previously.^{24,30} In brief, participants were able to read, write, and comprehend English; at least 18 years old; and attended a congregation within the two faith-based networks. As this study used a CBPR approach, data collection decisions were made in partnership with network staff, congregations, and researchers. The study team had a goal of recruiting 378 congregations, which was perceived as being burdensome for network staff if every congregation member was provided the survey. To make data collection manageable, purposive sampling was used to select 1% of each congregation to participate in the study. To reach this 1%, network staff worked with church leadership to identify at least one church leader and a diverse subset of church members who could provide a broad and representative perspective on each congregation's needs.³⁰ Surveys were self-administered using a paper-pencil or electronic version (Survey Monkey) and took approximately 20-30 minutes to complete.³⁰ The University of Memphis and Appalachian State University Institutional Review Boards (IRB) approved the University of Tennessee Health Science Center as the IRB of record.

Measure

The study examined data collected from *Mid-South Congregational Health Needs Assessment Survey (MSCHS)*.³⁰ Demographic information included: age (based on average age: <55 years old, ≥55 years old); sex assigned at birth (male, female); race/ethnicity [African American, Other race/ethnicity (ie, White, Hispanic/Latino, Asian, Native American, Multiracial, Other)]; marital status (married, not married), educational level (≤ high school, some college, ≥ college), and insurance status (yes, no). While interest was on individual-level needs, individuals were nested within churches. Thus, church-related data included: church role (leader, member), church size [small (≤100 members), medium (101-350 members), large (>351 members)], and church zip code [categorized as above or below the median household income (\$51, 657) of Memphis, TN].³¹

The 36-item health index was used to assess health-related needs and included perceived physical and mental health needs as well availability of different SDOHs. A description of the index with the full list of health-related needs has been previously published.³⁰ Respondents were provided a column where they could mark *all* needs they perceived existed within their congregation and the community where their congregation resided. Not placing a mark was interpreted as the participant did not perceive that item as a need.³⁰ In the index, diabetes encompassed all types (type 1, type 2, and gestational). Of the nine SDOH-related items,³⁰ the top three most reported needs were availability of affordable healthcare, healthy foods, and employment. These SDOHs were included in this analysis as they aligned with a natural break in the data and with needs identified in community assessments previously conducted in the Mid-South.²⁵⁻²⁷

Analysis

All data analyses were conducted using SAS Version 9.4. Descriptive statistics (mean and standard deviation for continuous variable; frequency and percentage for categorical variable) were used to summarize data. Excluded in the analysis were participants with missing data for sociodemographic variables (n = 46, 5.6%).

For the first aim, generalized linear mixed modeling (GLMM) was performed using PROC GLIMMIX to evaluate associations between sociodemographic variables (age, sex, race/ethnicity, educational level) and health-related needs (diabetes, affordable healthcare, healthy food, and employment). In the models, sociodemographic variables were included as predictors and health-related needs as outcomes. All models were adjusted for church size and church zip code. Since participants were nested within churches, two models were examined to assess the variability in reporting diabetes as a health concern across churches for each outcome. Model 1 included the fixed effects, or variability in responses from individual-level variables (ie, age, sex, race/ethnicity, educational level). Model 2 included the random effects, or variability in responses from individual-level as well as church-level (eg, church ID) variables. To compare the best fitting model, the Likelihood-Ratio Test (LRT) was performed using the difference between -2 Log Likelihood (-2LL) of model 1 and model 2; $P < .05$ indicated significant variability in responses across churches and the need to include the church-level variable in the model. Odds ratios (ORs) and 95% confidence intervals (CIs) of the best fitting model were used to examine associations between demographic variables and health-related needs.

For the second aim, GLMM was performed to evaluate associations between SDOH-related needs (affordable healthcare, healthy foods, and employment) as a categorical predictor, and reporting diabetes as a health concern or need in their community as a categorical outcome. Models were adjusted for age, sex, race/ethnicity, educational level, church size, and church zip code. Variations in reporting diabetes as a need across churches were examined by inspecting two models: Model 1 tested the fixed effects of level-1 variables (SDOH variables), and Model 2 tested the inclusion of a level-2 variable (church ID). The model with the best fit was again determined using the LRT method for assessing goodness-of-fit. ORs and 95% CIs of the best fitting model were inspected to examine the associations between each SDOH variable and reporting diabetes as a need.

Results

Sample Characteristics

Table 1 presents the study's sample characteristics. Of the 378 churches, 92 churches participated in the study (24% response rate), with 828 church leaders and members

Table 1. Sample Characteristics (N = 828).

Variable		M (SD) or n (%)
Age		54. 68 (15.4)
<55 years old		386 (46.6)
≥55 years old		442 (53.4)
Sex		
Women		575 (71.4)
Men		230 (28.6)
Race/Ethnicity		
African American		733 (90.5)
White		60 (7.4)
Other		17 (2.1)
Educational level		
≤ High school		142 (17.6)
Some college		334 (41.3)
College		332 (41.1)
Marital status		
Yes		367 (45.5)
Insurance status		
Yes		747 (95.9)
Church role		
Member		367 (48.3)
Leader		393 (51.7)
Church size	Range of respondents per church	
Small (≤100 members)	2-25	256 (30.9)
Medium (101 – 350 members)	3-42	263 (31.8)
Large (351 - 1000 members)	5-85	104 (12.6)
Mega (>1000 members)	5-82	194 (23.4)
Health-related needs		
Diabetes (yes)		475 (57.4)
Affordable healthcare (yes)		498 (60.1)
Healthy food access (yes)		474 (57.3)
Employment (yes)		463 (55.9)

M (SD) = mean (standard deviation) n (%) = frequency (percentage).

Other race/ethnic category included Hispanic/Latino, Asian, Native American, Multiracial, and other.

Some college included some college but no degree and 2-year degree.

College included 4-year degree and professional degree (professional, MD, Masters, PhD).

Church member included church member or regular attendee.

Church leader included pastor, pastor's spouse, associate pastor, non-clergy staff, volunteer or ministry lead, and other leadership roles.

Percentages do not add up to 100.0% due to missing data.

completing the survey. Participants were on average 55 years old ($SD = 15.40$) and there were more women than men (71% vs 29%). Most participants identified as African American (91%) and reported attending some college (41%) or graduating from a 4-year institution (41%). There were slightly more church leaders (52%) than church members (48%) and more participants who attended large-sized churches (36%) than small- and medium-sized churches (31% and 32%, respectively). Diabetes was the 3rd most commonly reported physical health-related need, with more than half of the respondents reporting diabetes as a concern in their community (57%). More than half of the participants also reported the following SDOH-related needs in their community:

availability of affordable healthcare (60%); healthy food (57%); and employment (56%).

The perception that diabetes is a health concern or need in their community was similar across age (≥55 years old [59%]; <55 years old [57%]) and education level (≤ high school [55%]; some college [58%]; college [59%]). There were less men compared to women (53% vs 60%, respectively), and more African Americans compared to individuals in the Other race/ethnic category (61% vs 27%, respectively) who reported diabetes as a need in their community. The majority of participants who selected diabetes as a health-related need also selected affordable healthcare (74%), healthy food (73%), and employment (76%) as SDOH-related needs (Table 2).

Table 2. Frequencies of Participants Who Selected Diabetes as a Need by Sociodemographic Variables and by Social Determinants of Health (N = 828).

	Diabetes	
	No n (%)	Yes n (%)
Sociodemographic variables		
Age ^a		
<55 years old	157(43)	205(57)
≥55 years old	182(41)	260(59)
Sex ^a		
Women	231(40)	344(60)
Men	108(47)	233(53)
Race/Ethnicity ^a		
African American	287(39)	446(61)
Other	56(73)	21(27)
Educational level ^a		
≤ High school	64(45)	78(55)
Some college	139(42)	195(58)
College	136(41)	196(59)
Social determinants of health		
Affordable health care		
No	222(67)	108(33)
Yes	131(26)	367(74)
Healthy food		
No	226(64)	128(36)
Yes	127(27)	347(73)
Employment		
No	242(66)	123(34)
Yes	111(24)	352(76)

^aDoes not equal to N = 828 due to missing data.

Aim 1: Associations Between Demographic Variables and Needs

The Likelihood-Ratio Test (LRT) was not statistically significant for diabetes ($P > .05$) indicating high agreement across churches that diabetes was a health need. Therefore, church ID was not included in analyses of associations between sociodemographic variables and reporting diabetes as a need. However, the LRT was statistically significant for availability of affordable healthcare ($P < .01$), healthy food ($P < .01$), and employment ($P < .01$) indicating variation across churches for these variables. Therefore, church ID was included in analyses of associations between sociodemographic variables and these SDOH-related needs.

When the odds of reporting diabetes as a health need or concern was examined across sociodemographic variable categories, only sex and race/ethnicity showed statistically significant differences between groups. Men had lower odds of reporting diabetes as a need compared to women (OR = .70; 95% CI = .50, .97). African Americans had greater odds of reporting diabetes as a need compared to individuals in the Other race/ethnic category (OR = 3.91; 95% CI = 2.20, 6.94).

When the odds of reporting availability of affordable healthcare, healthy food, and employment were examined, statistically significant differences were seen for all demographic variables. Compared to younger adults (<55 years), older adults (≥55 years) had lower odds of reporting employment as a need in their community (OR = .59; 95% CI = .42, .81). Men had lower odds of reporting availability of healthy food (OR = .64; 95% CI = .50, .90) and employment (OR = .70; 95% CI = .42, .81) as needs compared to women. African Americans had greater odds of reporting availability of employment as a need in their community compared to individuals in the Other race/ethnic category (OR = 2.60; 95% CI = 1.41, 4.81). Individuals with ≤ high school education had lower odds of reporting affordable healthcare (OR_{≤HS vs. some college} = .63; 95% CI 0.41, .99), availability of healthy food (OR_{≤HS vs. College} = .59; 95% CI = .38, .92), and employment (OR_{≤HS vs. some college} = .59; 95% CI = .38, .93) as needs compared to individuals with some college or a college degree (Table 3).

Aim 2: Associations Between SDOHs and Diabetes

As the LRT for variation in reporting diabetes as a need across churches was not statistically significant ($P > .05$), church ID was not included in analyses of the association between SDOHs and diabetes. After controlling for sociodemographic variables, individuals who reported affordable healthcare (OR = 2.54; 95% CI = 1.73, 3.73), healthy food (OR = 2.24; 95% CI = 1.55, 3.23), and employment (OR = 3.33; 95% CI = 2.30, 4.84) as needs had greater odds of also reporting diabetes as a need compared to individuals who did not report these SDOHs as needs (Table 4).

Discussion

Using a CBPR-driven congregational health needs assessment, this study found women and African Americans perceived diabetes to be an important need in their community. When associations between sociodemographic variables and SDOHs were examined, men were less likely to perceive healthy foods and employment as needs compared to women, while African Americans perceived availability of employment as a need in their community. Interestingly, individuals with lower educational attainment (high school education) were less likely to perceive SDOH-related needs (affordable healthcare, healthy food, employment) as a health concern compared to individuals with higher education (some college or college education). Congregations as well as their healthcare and community partners can use these findings to make decisions about where and to whom they market and design diabetes programs and policies.

In this study, diabetes was the 3rd most commonly reported physical health related-need, with approximately 57% reporting diabetes as a concern in their community. While diabetes was not reported as the top 1 health concern by our

Table 3. Associations Between Demographic Variables and Health-Related Needs (N = 828).

Variables	Diabetes ^a		Affordable Healthcare ^b		Healthy Food ^b		Employment ^b	
	OR	95% CI	OR	95% CI	OR	95% CI	OR	95% CI
Age								
≥55 years vs <55 years	1.01	.75, 1.38	.96	.69, 1.34	.89	.64, 1.23	.59	.42, .81
Sex								
Men vs Women	.70	.50, .97	.84	.60, 1.18	.64	.50, .90	.70	.50, .99
Race/Ethnicity								
African American vs Other	3.91	2.20, 6.94	1.77	.96, 3.25	1.57	.87, 2.84	2.60	1.41, 4.81
Educational level								
≤ HS vs Some college	.80	.53, 1.23	.63	.41, .99	.80	.52, 1.25	.59	.38, .93
≤ HS vs College	.67	.43, 1.03	.63	.40, 1.00	.59	.37, .92	.72	.46, 1.12
Some college vs College	.83	.60, 1.16	1.00	.71, 1.42	.73	.52, 1.02	1.20	.85, 1.70

OR = odds ratio; CI = confidence interval; Bolded values indicate significant associations.

Other race/ethnic category included White, Hispanic/Latino, Asian American, Native American, Multiracial, and Other.

HS = high school; Some college included some college but no degree or 2-year degree; College included ≥ 4-year degree.

Reference category: Age = <55 years; Sex = women; Race/ethnicity = Other; Educational Level = Some College (for ≤ HS vs Some college) and College (HS vs College; Some college vs College).

Models were adjusted for church size and church zip code.

^aLikelihood-Ratio Test indicated no significant random effect variation across churches; Model included fixed effects of level-I variables (ie, age, sex, race/ethnicity, educational level).

^bLikelihood-Ratio Test indicated significant random effect variation across churches; Model included fixed effect of level-I (ie, age, sex, race/ethnicity, educational level) and random effect of level-2 (ie, church ID) variables.

Table 4. Associations Between Social Determinants of Health Variables and Diabetes as a Need (N = 828).

Social determinants of health variables	OR	95% CI
Affordable healthcare	2.54	1.73, 3.72
Healthy food	2.24	1.55, 3.24
Employment	3.33	2.29, 4.84

OR = odds ratio; CI = confidence interval; Bolded values indicate significant associations.

Models were adjusted for age, sex, race/ethnicity, educational level, church size and church zip code.

Likelihood-ratio test for all models indicated no significant random effect variation across churches ($P < .05$); All models included fixed effects, level-I variables only (ie, affordable healthcare, healthy food, employment).

sample, it remains a significant health issue in the Mid-South, with local data indicating about 12% of the population living with diabetes.²⁵⁻²⁷ When differences in reporting diabetes as a health concern were examined across sociodemographic characteristics, men were less likely to report it as a concern compared to women, which was unexpected as national reports indicate higher rates of diagnosed and undiagnosed diabetes in men compared to women.¹ Given evidence of women participating in health promotion programs more often than men,³² it is possible that this is contributing to men's perception and awareness that diabetes is not a health need in their community. Diabetes programs should expand marketing efforts to appeal more men to participate in such initiatives and raise awareness about diabetes prevalence among men.

In addition, African Americans were more likely to perceive diabetes as a need compared to individuals in the 'Other' race/ethnic category. Efforts to address diabetes through faith-based health promotion programs have increased in recent years, especially in African American church communities.³³⁻³⁵ Despite these efforts, there continues to be a high prevalence of diabetes and related mortality among African Americans in the Mid-South and the U.S., which may contribute to their perception that diabetes is a significant health issue in their community.^{1,27,36} Given diabetes inequalities continue to impact African American communities in the Mid-South and the U.S., there is continued need for diabetes prevention and control programs that are tailored to the needs of African Americans. One way to tailor and enhance these programs is to acknowledge and begin addressing the SDOHs that are most pertinent to this community and align with diabetes.

When needs related to SDOH were examined, older adults (≥55 years) were less likely to report employment as a need compared to younger adults (<55 years). This finding was not surprising given the retirement age in the U.S. is as early as 62 years old³⁷ and the unemployment rate for adults aged 55 years or older is relatively lower compared to adults younger than 55 years old.³⁸ In addition, men also were less likely to perceive healthy food access and employment as needs compared to women. More women face inequalities in employment opportunities and labor market compared to men,³⁹ which has implications to one's socioeconomic status as well as other social needs including healthy food access.

Faith-based programs should examine strategies for empowering women in strengthening their career paths and economic stability (eg, classes related to leadership development, college prep, career planning, financial literacy program).

In this study, African Americans were more likely to report availability of employment as a health-related need in their community compared to individuals in the Other race/ethnic category. This finding is consistent with reports indicating a widening gap in unemployment rates between African Americans and Whites as well as other racial/ethnic minority groups in the Mid-South and the U.S.⁴⁰⁻⁴² While programs promoting economic stability including employment and job training have been implemented in African American churches,⁴³ forming partnerships and aligning efforts with public agencies may further enhance workforce training and development in the community.

In addition, individuals with less education were less likely to report affordable healthcare, healthy food access and employment as needs. This was unexpected as the association between lower educational attainment, decreased employment opportunities, and access to health-related resources (eg, healthy food and affordable healthcare) has been well documented.^{7,44} While the majority of the sample reported having at least some college education, low educational attainment remains a significant issue in the Mid-South.²⁷ Higher educational attainment has been shown to improve an individual's health literacy, which can help individuals navigate healthcare resources, enhance provider communication, adhere to treatment, and make informed decisions about their health.⁴⁵ Most health promotion programs implemented by faith-based organizations place significant emphasis on improving lifestyle behaviors (eg, healthy eating, promoting physical activity) and preventing chronic disease risk and prevalence (eg, diabetes).¹⁷ Based on findings from this study, faith-based programs should also include strategies for improving health literacy as well as examine ways they can assist individuals in accessing quality education (eg, GED classes, college enrollment assistance, financial aid and scholarship support) as part of addressing health comprehensively.²⁴

When associations between SDOH-related needs and diabetes were examined, individuals who perceived affordable healthcare as a need in their community also reported diabetes as a need. While most participants in the current study reported having health insurance (96%), access to affordable and quality healthcare continues to be an important public health issue in the Mid-South.²⁵ Many individuals experience other healthcare-related costs (eg, insurance cost, out-of-pocket, copayment),⁴⁶ which greatly impacts one's ability to receive optimal diabetes screening and care.⁷ To date, Mississippi and Tennessee are among the ten states who have not accepted federal funding to expand Medicaid and extend it to adults under 65 years old with incomes up to 138% of the federal poverty level.⁴⁷ Historically, the assassination of Dr Martin

Luther King Jr in Memphis, TN stemmed from his fight for economic equality in the civil rights era.⁴⁸ However, inequities persist and continue to impact access to quality healthcare, especially among African Americans in the Mid-South.⁴⁹ Diabetes education programs should include educational content to increase awareness of recent healthcare-related (eg, Medicare and Medicaid) policies as well as resources that may encourage one to advocate to their legislators about important policy changes related to healthcare access.

In addition, participants who perceived healthy food access as a need also reported diabetes as a need. Lower access to healthy food has been linked with increased diabetes incidence,⁵⁰ while living in areas with more supermarkets has been linked with reduced diabetes risk.⁵¹ Lack of access to healthy food has been reported as an important public health issue in the Mid-South, with residents experiencing food insecurity at about 1.51 to 2.22 times greater than the general U.S. population.²⁷ Prior interventions indicate access to food banks and food pantries improved lifestyle behaviors (eg, increased fruit and vegetable consumption), psychosocial (eg, greater self-efficacy), and clinical outcomes (eg, lower hemoglobin A1C, body mass index) in food insecure people with diabetes.^{52,53} In the Mid-South, many food pantries operate in churches and other faith-based institutions [eg, Metropolitan Inter-Faith Association (MIFA), Memphis Interfaith Coalition for Action and Hope (MICAHA), and other catholic charities].⁵⁴ Thus, increasing partnerships with local healthcare systems offering food pantry programs⁵⁵ and rigorously evaluating existing strategies using evidence-based frameworks [eg, RE-AIM Framework⁵⁶] may help further mobilize these assets and expand the reach to more food insecure communities.

In this study, individuals who reported employment as an important health-related need also reported diabetes as a health concern in their community. While unemployment rates in the Mid-South have declined in the recent decade,⁴⁰ it was reported at about 3.4% (ie, western Tennessee) to 3.9% (ie, northern Mississippi) in 2019, which was similar to the overall U.S. rate of unemployment (3.4%).²⁷ A meta-analysis reported increased odds of both prediabetes and diabetes among individuals experiencing unemployment.⁵⁷ As part of addressing health comprehensively, faith-based programs should partner with local public agencies (eg, Memphis Public Libraries) offering programs designed maximize one's employment opportunities including career counseling and expert advice from employment specialists.⁵⁸

Limitations and Strengths

There were several limitations to the study. The current sample included a large proportion of African Americans and women, and findings may not be generalizable to all church attendees. However, it is notable that the current sample is representative of church attendees in the U.S.^{59,60} The study goal was to collect survey data from 378 congregations, yet only 92 churches participated in the study (24% response rate).

While this project was largely embedded into the work responsibilities of the network staff administering the surveys, other responsibilities within their organization may have impacted their ability to reach more churches. However, network staff reached out to churches they knew would be interested or could use the information, instead of only reaching out to churches that may have the funds or resources to implement programs. While there were plans to continue efforts to collect congregational health needs data after the first year, the COVID-19 pandemic began and a shift in priorities within the healthcare system led to an end in the study before collecting more data to increase the sample size. Additionally, among churches that preferred taking paper-pencil surveys, practical challenges (scheduling constraints with other church events, limited space to administer the surveys) may have impacted their willingness to participate, especially in small-sized churches.⁶¹

In addition, there were a small number of respondents in the 'Other' racial/ethnic category, making it difficult to interpret differences in needs in these groups. The study data were also collected before the COVID-19 pandemic, and it is unknown how the pandemic impacted the perception of diabetes, affordable healthcare, healthy food, and employment as key health issues. In the study, diabetes was defined more broadly rather than by a specific type (ie, type 1, type 2, gestational). While all types of diabetes are characterized by elevated blood glucose levels, type 1 and gestational diabetes are more medically complex than type 2 diabetes.⁶² There are also nuances related to prevention and management for each type of diabetes, which could have impacted one's perception of diabetes as a health-related need in their community. The current analyses only examined three of the 9 SDOH-related needs listed in the MSCHS health index. However, the SDOH-related needs examined in the current study align with local community health needs assessments in the Mid-South.²⁵⁻²⁷ Focusing on these SDOH-related needs may assist organizations in prioritizing resources and developing tailored faith-based health promotion programs to address these needs.

Despite these limitations, several strengths have been identified. To our knowledge, this is the first study to examine associations between diabetes and SDOH-related needs using data from a congregational health needs assessment. Our multisector partnership and CBPR approach enhanced our ability to develop a comprehensive congregational health needs assessment tool.³⁰ Findings have important implications when designing or adapting diabetes prevention and management programs in congregations and the communities they serve.

Conclusions

Despite medical advances and prevention efforts, certain communities continue to be disproportionately burdened by diabetes, which is often compounded by SDOH that negatively impact health outcomes. Examining potential strategies that could help buffer the impact of negative SDOH (ie, lack of

access to resources) against poor diabetes outcomes is an important public health priority. Findings from the current study provide insight into populations reporting diabetes, and access to affordable healthcare, healthy food, and employment as health-related needs in the Mid-South, a geographic region in the U.S. experiencing high rates of diabetes as well as these SDOHs.^{3,25-27,63} Findings reinforce that employment opportunities, access to education, affordable healthcare and healthy food are important public health considerations, which can help in the adaptation of existing evidence-based diabetes programs and interventions [eg, National Diabetes Prevention Program⁶⁴]. CBPR approaches involve identifying and building strengths within communities, allowing partnerships to mobilize social and environmental processes that influence health.¹⁶ Future studies should evaluate effective CBPR strategies to enhance multisector partnerships and merge efforts between FBOs, healthcare institutions, and public agencies to help build multilevel programs to address the underlying SDOH impacting the disproportionate burden of diabetes in the U.S.

So What? (Implications for Health Promotion Practitioners and Researchers)

What is Already Known on This Topic?

Diabetes disproportionately impacts certain populations and geographic regions in the U.S. including the Mid-South, which is often compounded by social determinants of health that negatively impact health. However, the link between diabetes and SDOH has never been evaluated using a congregational health needs assessment.

What Does This Article Add?

Study findings reinforce that employment opportunities and access to quality education, affordable health care, and healthy foods are important public health considerations, especially when developing or adapting diabetes health promotion programs.

What are the Implications for Health Promotion Practice or Research?

Health promotion practitioners and researchers should examine strategies to merge efforts between healthcare and faith-based organizations to address the SDOH impacting diabetes outcomes.

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Author Contributions

ERNSD conceived of the study, analyzed the data, and assisted with manuscript writing and editing; NTW, YJ assisted with analysis, manuscript writing and editing; LCP, TWP, FR, JL, BC, LM, SM assisted with manuscript writing and editing; BEH conceived of the study, assisted with analysis and manuscript writing and editing.

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